1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Colour)
  - Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
  - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

Data Loading: Imported the dataset using pandas and used df.head() for initial table check.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

Initial Exploration: Used df.info () to check structure and df.describe () for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Checking of Null Values and Replacing it with median according to respective category: The aim is to replace null ratings with median value but just changing it to median is not a good idea because each category has different median and may introducing bias, hence we will use median review rating of each category and replace it with missing values according to category.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating              37
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
dtype: int64
```

Column Standardization: Each column name converted into snake case for better readability and documentation.

Converting categorical data into numerical data: Age group divided into 4 equal groups Youn adult, Adult, Middle aged, and Senior. Similarly, purchase frequency is given in the form of strings which is converted into numerical data.

A separate column is created for both of them using the following code-

```
#create a column age_group
labels = ['Young adult', 'adult', 'middle-aged', 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
#divides age into 4 equal groups based on number of data and name these group labels as labels defined above
```

```
frequency_mapping = {
'Fortnightly':14,
'Weekly':7,
'Monthly':30,
'Quarterly':90,
'Bi-Weekly':14,
'Annually':365,
'Every 3 Months':90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

Data consistency check: Verified if discount_applied and promo_code_used

were redundant; dropped promo_code_used.

Final Step: Database integrated to PostgreSQL.

4. Data Analysis using SQL:

Q1. What is the total revenue generated by male vs. female customers?

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Q2. Which customers used a discount but still spent more than the average purchase amount?

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Q3. Which are the top 5 products with the highest average review rating?

	item_purchased text	avg_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

	shipping_type text	avg_amount numeric
1	Standard	58.46
2	Express	60.48

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	customer_count bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

Q6. Which 5 products have the highest percentage of purchases with discounts applied

	item_purchased text	percent_purchase numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Q7. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	subscription_status text	repeat_purchase bigint
1	No	2847
2	Yes	1053

Q8. What is the revenue contribution of each age group?

	age_group text	revenue numeric
1	Young adult	62143
2	middle-aged	59197
3	adult	55978
4	senior	55763

Q9. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

	customer_segment text	total_count bigint
1	Loyal	3116
2	Returning	701
3	New	83

5. Power BI Dashboard: